

- **Extensibility:** Your AI system is unique. Open source tools let you extend instrumentation to custom retrieval pipelines, proprietary vector databases, and in-house orchestration frameworks.
- **Cost governance:** Enterprise-grade observability platforms charge per seat, per event, or per GB. At LLM token volumes, proprietary observability costs can rival the inference spend itself. Open source shifts that cost to compute.

The most sophisticated AI observability stacks at hyperscalers and leading enterprises are open source at their core — OpenTelemetry for the data plane, Grafana for visualisation, and purpose-built LLM-native tools for the semantic layer.

The open source observability landscape for AI

The tooling landscape splits into two layers: the foundational observability infrastructure — largely inherited from cloud-native — and the LLM-native semantic layer built specifically for AI workloads.

Foundational layer: Cloud-native observability:

OpenTelemetry (OTel) is the gravitational centre — the CNCF standard for telemetry and a vendor-neutral SDK for capturing traces, metrics, and logs across any language or framework. For AI observability, OTel serves as the transport layer: every tool in the LLM-native layer either exports to OTel or is built on top of it. Prometheus handles metrics storage and alerting. Grafana provides unified dashboards. Loki aggregates logs. Jaeger and Tempo handle distributed traces.

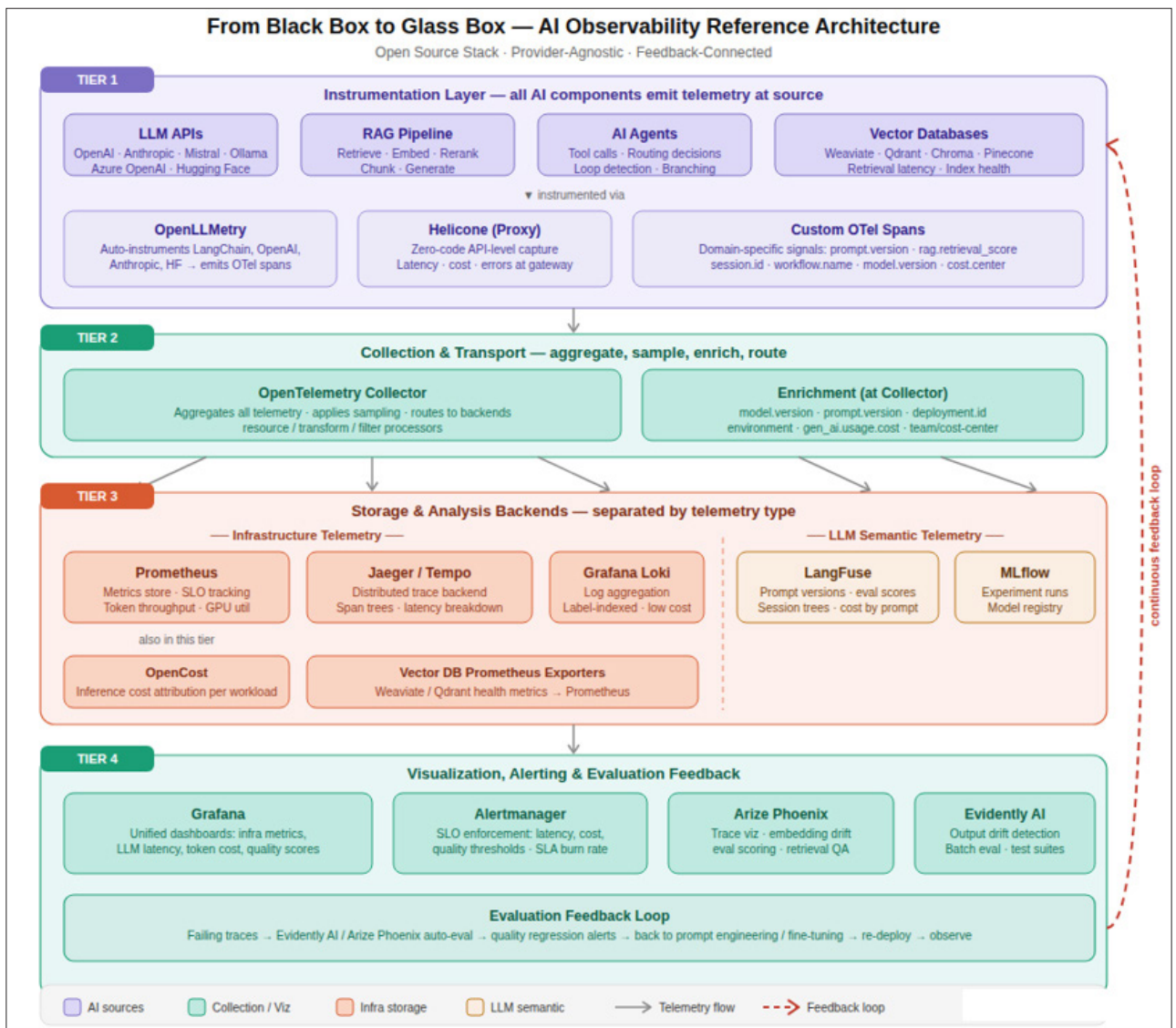


Figure 1: Open source AI observability reference architecture — four-tier stack with continuous evaluation feedback loop